

Work Breakdown Structure (WBS)

Date: October 2025

Sub-Project Name: Member Registration & Authentication System

1.0 Project Management & Planning

- 1.1 Project Initiation
- 1.2 Requirements Gathering & Analysis
- 1.3 Creation of Business Case, Charter, Scope Statement
- 1.4 Schedule, Budget, and Risk Monitoring
- 1.5 Weekly Status Reporting
- 1.6 Final Documentation & Lessons Learned
- 1.7 Project Close-out & Submission

2.0 System Design & Development

2.1 System Architecture & Database Design

- 2.1.1 Firestore “members” Collection Structure
- 2.1.2 Define Required Fields (name, email, phone, DOB, PIN, card number)
- 2.1.3 Firebase Configuration Setup (db, rules, initialization)

2.2 Registration Module Development

- 2.2.1 Frontend Registration Form UI (React.js)
- 2.2.2 Input Validation (required fields, PIN = 4 digits, phone = 10 digits)
- 2.2.3 8-Digit Library Card Number Generator Function
- 2.2.4 Store New Member in Firestore (addDoc)
- 2.2.5 Registration Success Message Handling

2.3 Authentication Module Development

- 2.3.1 Login Page UI (React.js)
- 2.3.2 Input Validation (card length, PIN length)
- 2.3.3 Firestore Query for Authentication
- 2.3.4 Error Handling (incorrect login, missing inputs)
- 2.3.5 Redirect to Home Page on Successful Login

2.4 Home Page Module

- 2.4.1 Display User Information After Login
- 2.4.2 Implement Logout Functionality (clear localStorage)

3.0 Testing & Quality Assurance

3.1 Functional Testing

- 3.1.1 UC-1 Registration Tests
- 3.1.2 UC-2 Login Tests
- 3.1.3 UC-4 Logout Tests
- 3.1.4 Card Number Generation Testing
- 3.1.5 Input Validation Error Testing

3.2 Database Testing

- 3.2.1 Firestore Write & Read Tests
- 3.2.2 Duplicate Card Number Collision Check (basic check only)

3.3 UI/UX Testing

- 3.3.1 Form Behavior Testing
- 3.3.2 Performance Check (page load under 2–5 seconds)

4.0 Deployment & Finalization

- 4.0.1 Firebase Hosting Deployment
- 4.0.2 Firestore Rules Configuration
- 4.0.3 Deployment Verification Checklist
- 4.0.4 Bug Fixes and Final Adjustments
- 4.0.5 Final Packaging for Submission